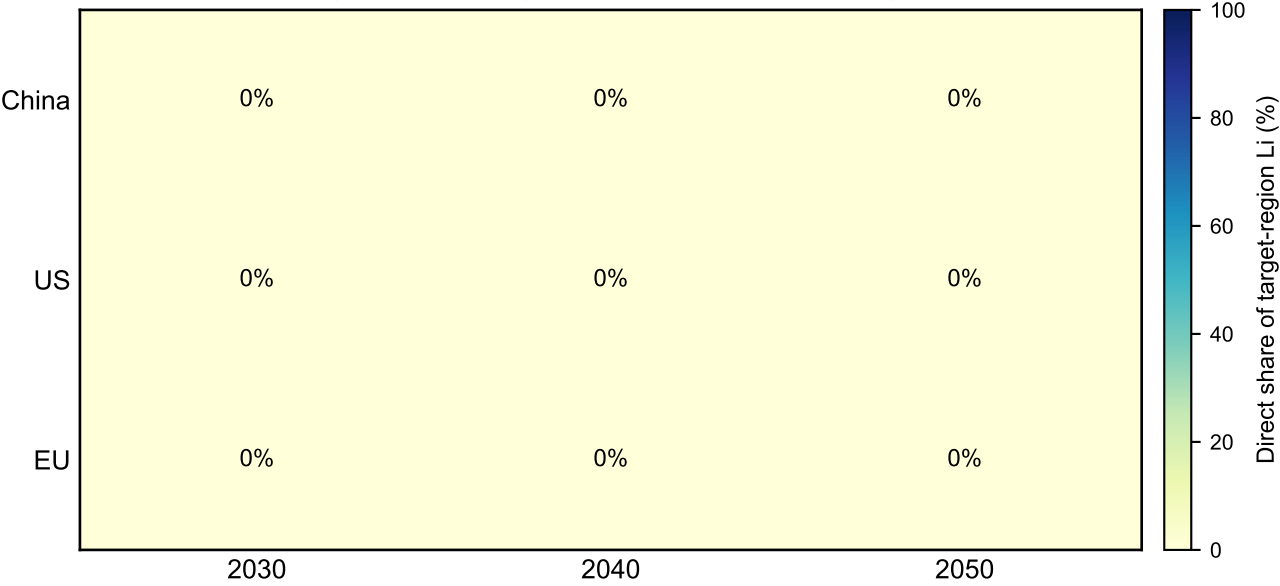


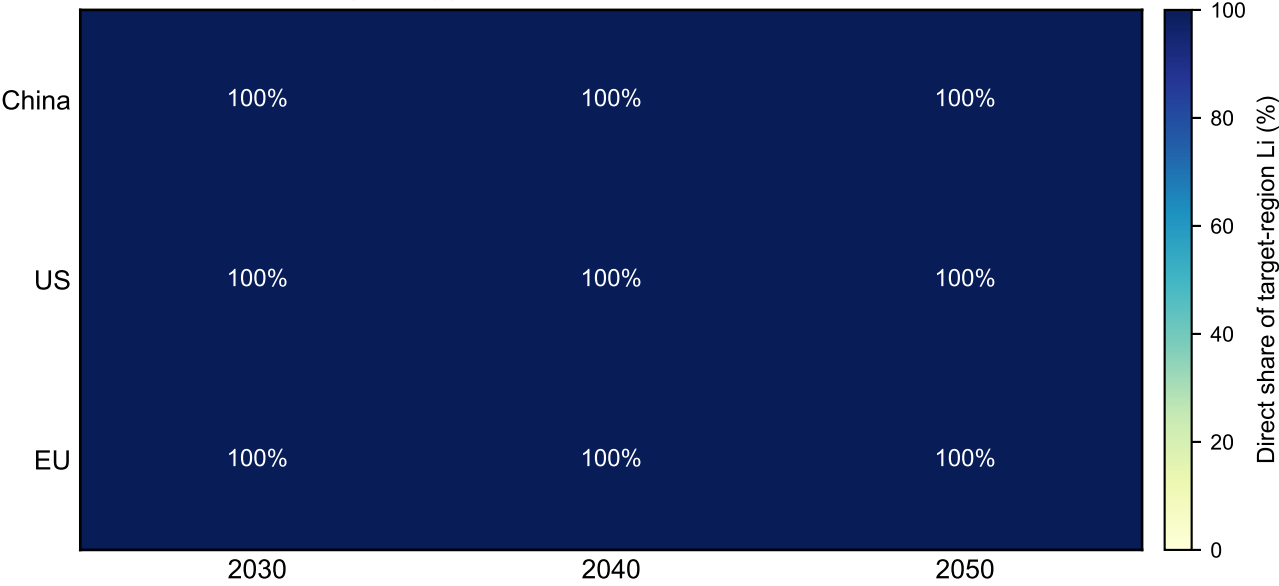
Technology substitution thresholds for regional secondary-lithium access

2030 2040 2050

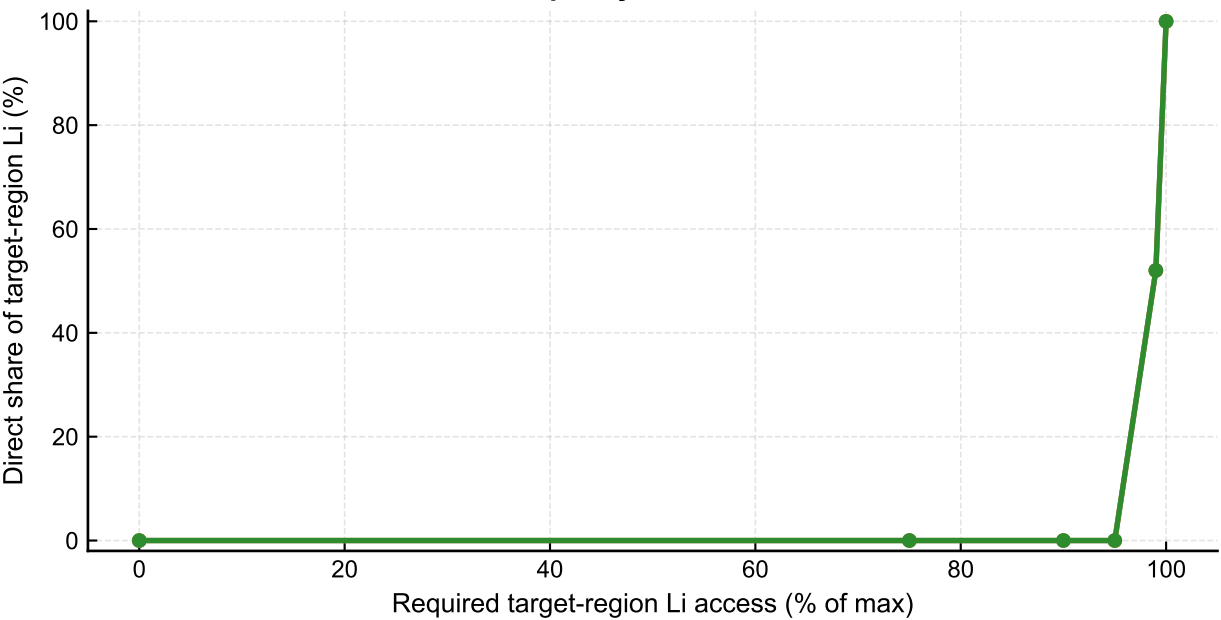
A. 95% access target remains Hydro-dominated across regions



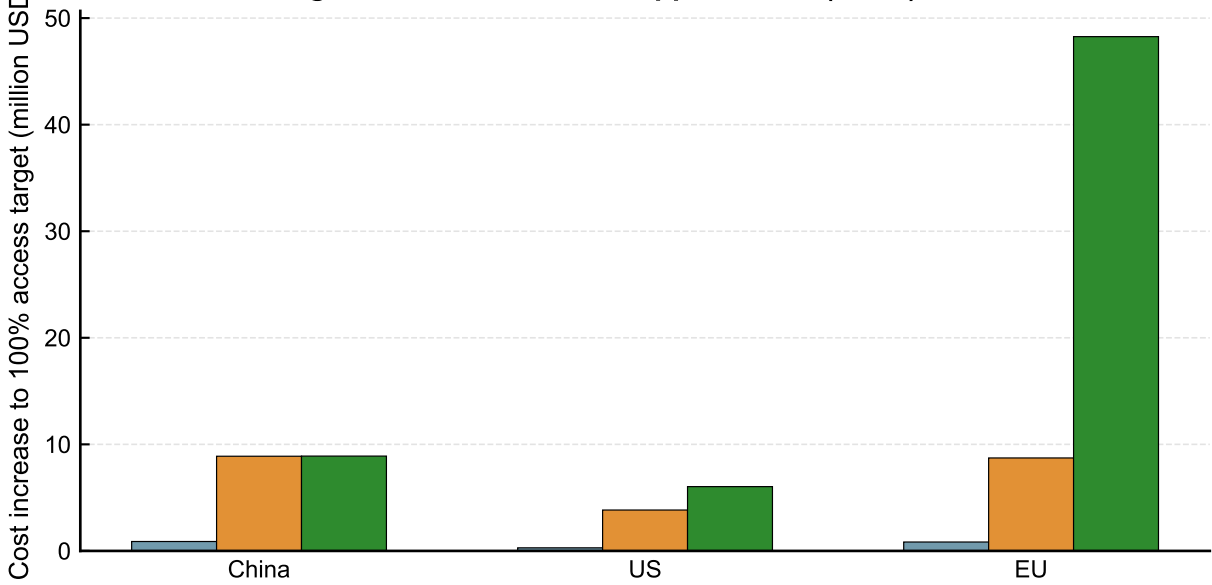
B. 100% access target triggers Direct-dominated solutions



C. EU Direct frontier under Strict policy



D. Cost of forcing the Direct-dominated upper bound (Strict)



China, US, and EU aggregate results under strict policy. Soft regional access targets show a threshold response: Hydro satisfies most sub-maximum access requirements, while Direct appears at the upper bound.